

Question 2 SQL Queries

A. Write a query using MySQL to return all countries included in the World database.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
3 -- Purpose: The purpose of the query is to return all the Names of countries in the database.
4 -- Tables Involved: Country table in the World database.
5 -- Filtering Logic: No filtering was used or needed for this task.
6 -- The query satisfies the requirement because it returns all countries in the World database as required.
7 --
8 SELECT Name AS Country_Name
9 FROM world.country
10
11
12
```

The Result Grid shows the following data:

Country_Name
Aruba
Afghanistan
Angola
Anguilla
Albania
Andorra
Netherlands Antilles

The Information panel shows the table structure for 'country':

Columns:	char(3)
Code	char(3)
Name	char(52)
Continent	enum(7)
Region	enum(7)
SurfaceArea	decimal(26)
IndepYear	smallint
LifeExpectancy	float

The Output panel shows the execution results:

#	Time	Action	Message	Duration / Fetch
5346	22:46:52	SET autocommit=@old_autocommit	0 row(s) affected	0.000 sec
5347	22:54:05	select * from country limit 20	20 row(s) returned	0.000 sec / 0.000 sec
5348	22:55:50	SELECT Name as Country_Name from world.country limit 20	20 row(s) returned	0.000 sec / 0.000 sec
5349	22:56:12	SELECT Name as Country_Name from world.country LIMIT 0, 1000	239 row(s) returned	0.000 sec / 0.000 sec
5350	23:40:06	SELECT Name as Country_Name from world.country LIMIT 0, 1000	239 row(s) returned	0.000 sec / 0.000 sec
5351	23:42:14	SELECT distinct Name AS Country_Name FROM world.country LIMIT 0, 1000	239 row(s) returned	0.000 sec / 0.000 sec

B. Write a query using MySQL that returns all countries in Europe.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
4 -- Tables Involved: Country table in the World database.
5 -- Filtering Logic: To return all countries in Europe only, we have to use 'where' to filter
6 -- and the operator '=' to assign a value to the variable we are filtering with. In this case, 'Continent'.
7 -- The query satisfies the requirement because it successfully filters the continent by the required value and returns
8 -- all the countries in Europe.
9 --
10 SELECT Name AS Country_Name
11 FROM world.country
12 WHERE Continent = 'Europe'
13
```

The Result Grid shows the following data:

Country_Name
Albania
Andorra
Austria
Belgium
Bulgaria
Bosnia and Herzegovina
Belarus

The Information panel shows the table structure for 'country' (same as above).

The Output panel shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	23:55:05	SELECT Name AS Country_Name FROM world.country WHERE Continent = 'Europe' LIMIT 0, 1000	46 row(s) returned	0.000 sec / 0.000 sec

C. Write a query using MySQL to return all countries that speaks English as an official language with percentage greater than or equal to 70.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

SELECT Name AS Country_Name, Language, Percentage, Isofficial
FROM world.country c
JOIN countrylanguage cl ON c.Code = cl.CountryCode
WHERE Language = 'English'
AND Isofficial = 'T'
AND Percentage >= 70;

```

The Result Grid shows the following data:

Country_Name	Language	Percentage	Isofficial
Australia	English	81.2	T
Bermuda	English	100.0	T
United Kingdom	English	97.3	T
Gibraltar	English	88.9	T
Ireland	English	98.4	T
New Zealand	English	87.0	T
United States	English	86.2	T

The bottom panel shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
11	00:23:18	select "from country where name like "U.S." or name like "U.S.A." LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
12	00:23:27	select "from country LIMIT 0, 1000	239 row(s) returned	0.000 sec / 0.000 sec
13	00:23:56	select "from country where name like "U.S." or name like "U.S.A." LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
14	00:26:21	SELECT Name AS Country_Name, Language, Percentage, Isofficial FROM world.country c JOIN c...	8 row(s) returned	0.000 sec / 0.000 sec
15	08:41:20	SELECT Name AS Country_Name, Language, Percentage, Isofficial FROM world.country c JOIN c...	8 row(s) returned	0.000 sec / 0.000 sec

D. Write a query using MySQL to return all countries that received their independence in the 18th century.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

SELECT Name AS Country_Name, IndepYear
FROM world.country
WHERE IndepYear >= 1701 and IndepYear <= 1800;

```

The Result Grid shows the following data:

Country_Name	IndepYear
Nepal	1769
United States	1776

The bottom panel shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
15	08:41:20	SELECT Name AS Country_Name, Language, Percentage, Isofficial FROM world.country c JOIN c...	8 row(s) returned	0.000 sec / 0.000 sec
16	08:53:32	SELECT "FROM world.country LIMIT 0, 1000	239 row(s) returned	0.016 sec / 0.000 sec
17	10:40:34	SELECT Name AS Country_Name FROM world.country c WHERE IndepYear >= 1701 and IndepY...	2 row(s) returned	0.000 sec / 0.000 sec
18	10:40:52	SELECT Name AS Country_Name, IndepYear FROM world.country c WHERE IndepYear >= 1701 ...	2 row(s) returned	0.000 sec / 0.000 sec
19	10:51:52	SELECT Name AS Country_Name, IndepYear FROM world.country c WHERE IndepYear >= 1701 ...	2 row(s) returned	0.016 sec / 0.000 sec

E. Write a query using MySQL that returns the following dataset.

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```
SELECT Name, CountryCode, District
FROM world.city
WHERE CountryCode = 'IND'
and District in ('Delhi', 'Punjab')
```

The Results Grid displays the following data:

Name	CountryCode	District
Delhi	IND	Delhi
Ludhiana	IND	Punjab
Amritsar	IND	Punjab
Jalandhar (Jullundur)	IND	Punjab
New Delhi	IND	Delhi
Patiala	IND	Punjab
Bhatinda (Bathinda)	IND	Punjab

The Action Output pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
19	10:51:52	SELECT Name AS Country_Name, IndepYear FROM world.country c WHERE IndepYear >= 1701 ...	2 row(s) returned	0.016 sec / 0.000 sec
20	10:53:29	SELECT Name, Code as CountryCode, District FROM world.city WHERE Code = 'IND' LIMIT 0, 1000	Error Code: 1054. Unknown column 'Code' in 'field list'	0.000 sec
21	10:54:23	SELECT * FROM world.city LIMIT 0, 1000	1000 row(s) returned	0.016 sec / 0.000 sec
22	10:54:54	SELECT Name, CountryCode, District FROM world.city WHERE CountryCode = 'IND' LIMIT 0, 1000	341 row(s) returned	0.000 sec / 0.000 sec
23	10:55:52	SELECT Name, CountryCode, District FROM world.city WHERE CountryCode = 'IND' and District in ...	12 row(s) returned	0.000 sec / 0.000 sec